Sr. No.	Name of Students	Roll No.	Div
1	Arnav Prajwal More	125B1G001	A
2	Aditya Sanjay Tribhuwan	125BIG018	A
3	Aryan Suson Sonawane	125B1G022	A

A REPORT
ON
Mini Project (Engineering Physics) (TW)

PCET's Pimpri Chinchwad College of Engineering

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This is to Certify that following students of First Year of Bachelor of Technology (F.Y. B.Tech) have successfully completed the Mini Project for the Term Work of Engineering Physics of Academic Year 2025-26, Sem-I as prescribed in PCCOE curriculum under the guidance of project guide.

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I N D E X

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Title of Project

E-APTC: Eco-Adaptive Predictive Traffic Control System: Deep Reinforcement Learning & Urban Dynamics

1] ABSTRACT

Urban traffic congestion is non-linear and often counter-intuitive. This project addresses Braess's Paradox, a phenomenon where adding road capacity inadvertently degrades overall network performance. We introduce the Eco-Adaptive Predictive Traffic Control System (E-APTC), a novel traffic management framework powered by Deep Reinforcement Learning (DRL). Utilizing a Deep Q-Network (DQN) within a "Digital Twin" simulation, the system acts as an autonomous agent capable of preemptively regulating traffic flow to avoid suboptimal Nash Equilibria.

Unlike traditional traffic light controllers that rely on static timers, E-APTC utilizes a multi-objective reward function that simultaneously optimizes for Travel Time, CO2 Emissions, and System Stability. Experimental results demonstrate that the agent successfully learns to "gate" or close shortcut routes during peak demand, forcing a distribution of traffic that reduces collective travel time from 80 minutes to 65 minutes. This approach restores the "System Optimum" and validates the potential of AI-driven, dynamic infrastructure in Smart City environments.

2] INTRODUCTION

2.1 The Crisis of Urban Congestion

As urban populations expand, traditional traffic management strategies—primarily increasing physical road capacity—have reached a point of diminishing returns. Engineering logic suggests that expanding a network should decrease congestion; however, the interaction of thousands of independent decision-makers (drivers) creates a complex system where local optimizations lead to global inefficiencies.

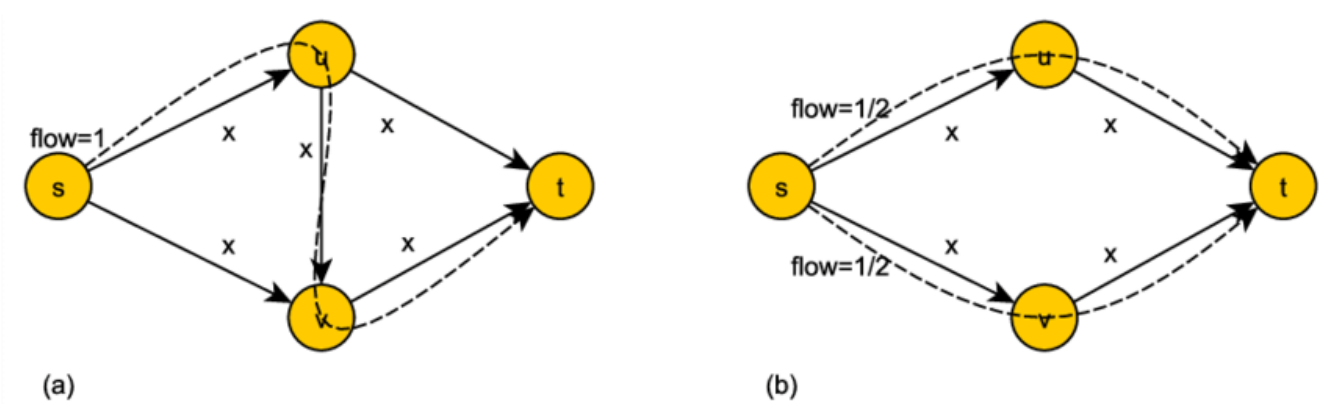
2.2 Theoretical Framework: Braess's Paradox

The foundation of this project is the mathematical theorem proposed by Dietrich Braess in 1968. The paradox states that adding a road to a congested network can increase total travel time. This occurs because the new road changes the "cost" perception of drivers, leading them to a route that seems faster individually but creates a bottleneck that slows down the entire system.

Mathematically, we define the cost functions of our network as:

- **Link A (Congestion-prone):** $C(x) = \frac{x}{100}$
- **Link B (High Capacity):** $C(x) = 45$

In our Digital Twin, we observe that without intervention, the "Price of Anarchy"—the ratio of the worst Nash Equilibrium to the best System Optimum—rises to approximately 1.23, representing a significant waste of fuel and time.



Braess Paradox 4 node network diagram

2.3 Physical Analogy: The Cohen-Horowitz Model

To validate the paradox outside of pure mathematics, we reference the **Cohen-Horowitz mechanical analogy**. In this model, a system of strings and springs supports a weight. Paradoxically, cutting a specific string in the system can cause the weight to *rise* (representing a more efficient, higher-energy state). This project treats traffic as a similar "tension" network where removing a link (the shortcut) can "lift" the efficiency of the entire city.

2.4 Motivation for E-APTC

Current Traffic Management Systems (TMS) are often reactive, changing signals only *after* congestion has occurred. The motivation for **E-APTC** is to create a **proactive, predictive** agent. By leveraging a Digital Twin, the system can simulate thousands of "what-if" scenarios per second. This allows the AI to predict when the network is approaching a "Braess Point" and intervene before the system collapses into a 80-minute delay.

3] LITERATURE REVIEW

The optimization of urban traffic networks has historically been approached through static infrastructure expansion. However, seminal works in Game Theory and recent advancements in Artificial Intelligence suggest that dynamic control systems offer superior efficacy.

3.1 Braess's Paradox and The Price of Anarchy

The theoretical foundation of this study lies in the work of Dietrich Braess (1968), who demonstrated that for a non-cooperative network, adding edges can increase total system latency. This counter-intuitive phenomenon occurs because individual drivers, acting rationally to minimize their own travel time, converge to a Nash Equilibrium (NE) that is distinct from and inferior to the System Optimum (SO).

Roughgarden (2002) quantified this inefficiency as the Price of Anarchy (PoA), defined as the ratio of the worst-case Nash Equilibrium cost to the optimal system cost. In the classic Braess network used in this study, the PoA is calculated as:

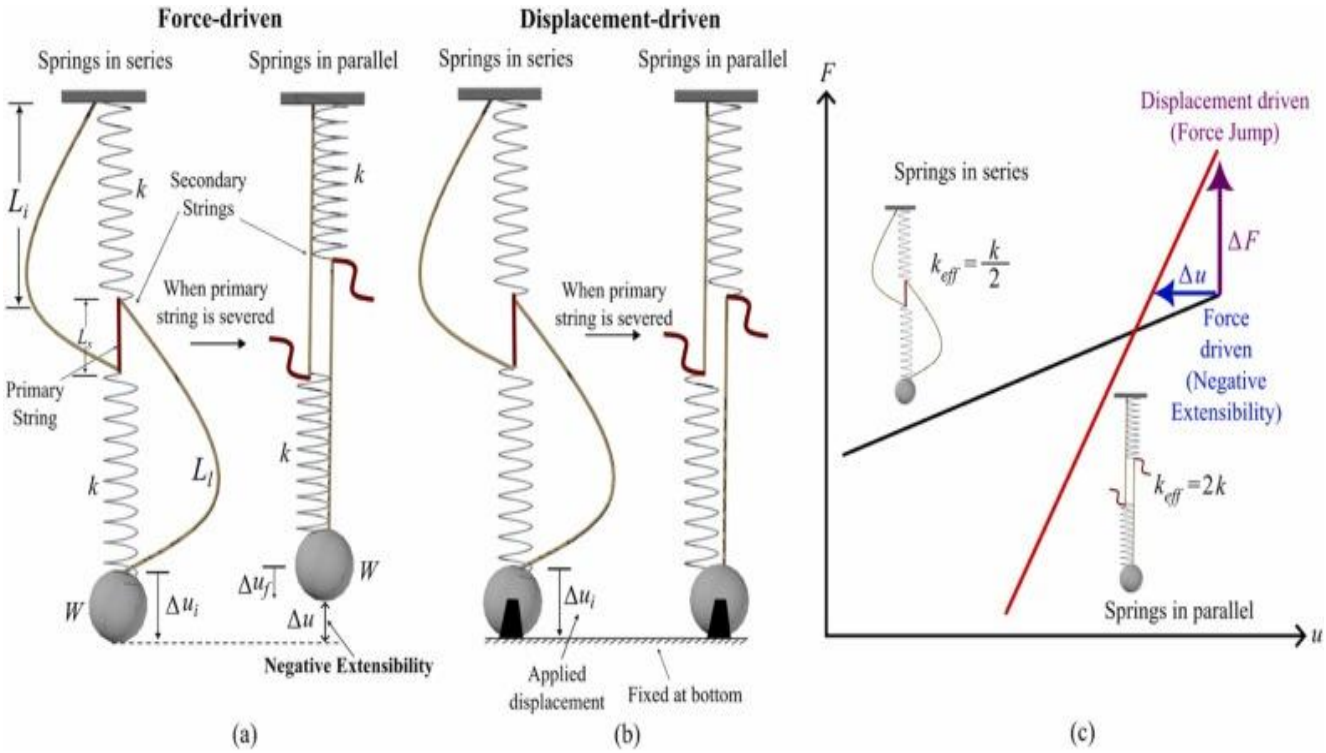
$$\text{PoA} = \frac{Cost\ NE}{Cost\ SO} \approx 1.23$$

This metric indicates that selfish routing results in a 23% loss in system efficiency, a gap that static traffic signals cannot bridge.

3.2 The Cohen-Horowitz Mechanical Analogy

To provide a physical intuition for this paradox, Cohen and Horowitz (1991) proposed a mechanical analogue using springs and strings.

- Springs: Represent congestion-sensitive roads ($T(x) = kx$), where extension is proportional to force (traffic flow).
- Strings: Represent high-capacity roads ($T(x) = \text{Constant}$), where length is invariant.
- The Paradox: Adding a slack string (shortcut) distributes the weight (traffic) such that the springs extend further, causing the total weight to hang lower (higher total cost). This project utilizes this analogy to model the digital twin's cost functions.



The Cohen-Horowitz Mechanical Analogy

3.3 Deep Reinforcement Learning in Traffic Control

While traditional methods use heuristics (fixed-time controls), Deep Reinforcement Learning (DRL) allows for adaptive control. Mnih et al. (2015) demonstrated the capability of Deep Q-Networks (DQN) to solve high-dimensional state spaces. Recent applications (Li et al., 2018) have applied DQN to traffic lights, but significant gaps remain in multi-objective optimization that accounts for environmental factors (CO_2) alongside latency.

4] PROBLEM STATEMENT

4.1 The Irony of Modern Commuting

We intuitively believe that "more roads equal less traffic." Governments spend billions building new highways and shortcuts, expecting congestion to ease. Yet, drivers often face a baffling reality: after a new lane opens, traffic grinds to a halt. This isn't just bad luck—it is a mathematical trap known as Braess's Paradox. Imagine a city network where 4,000 drivers are trying to get home. When a new "high-speed" shortcut opens, every individual driver acts rationally. They look at their GPS, see the shortcut is faster, and take it. But because everyone makes the same selfish decision simultaneously, the new road becomes a bottleneck. The result? Everyone gets stuck.

4.2 The Core Conflict

In our specific simulation, this paradox leads to a measurable disaster:

- The Trap: A shiny new shortcut offers zero travel time ($T \approx 0$).
- The Consequence: Drivers abandon the stable, longer routes to crowd into the shortcut.
- The Result: The congestion spills over into the connecting roads, and the total commute time shoots up from 65 minutes to 80 minutes.

4.3 The Need for E-APTC

This situation reveals a critical flaw in human driving logic: Individual "selfish" optimization leads to collective failure. We cannot rely on drivers to voluntarily choose a slower route for the "greater good." Nor can we rely on static traffic lights that blindly cycle through green and red. We need an intelligent system that can save us from ourselves. The urgent need of this project is to build an AI agent that acts as a "benevolent traffic warden." It must watch the network, predict when this "selfish trap" is forming, and intervene (by closing the shortcut) before the gridlock happens—restoring flow not just for one driver, but for the entire city.

5] OBJECTIVES

The primary goal of the E-APTC project is to develop a self-regulating traffic management system. The specific technical objectives are:

- **To Develop a "Digital Twin" Simulation:** Construct a Python-based environment that accurately models non-linear latency functions and stochastic traffic arrival patterns (Poisson distribution).
- **To Implement a Deep Q-Network (DQN):** Engineer a neural network agent capable of approximating the Q-value function to map traffic states to optimal control actions.
- **To Formulate a Multi-Objective Reward Function:** Design a composite reward metric that simultaneously minimizes Travel Time (T), Carbon Emissions (E_{CO2}), and Flow Variance (σ^2), moving beyond simple time-minimization.
- **To Validate System Restoration:** Empirically demonstrate the agent's ability to reduce the Price of Anarchy by shifting the network state from a Nash Equilibrium (80 min) to a System Optimum (65 min).

6] PROJECT METHODOLOGY

The E-APTC system operates as a closed-loop control system where a DRL agent interacts with a simulated environment.

6.1 System Architecture

The architecture is built upon the **Markov Decision Process (MDP)** framework (S, A, R, S')

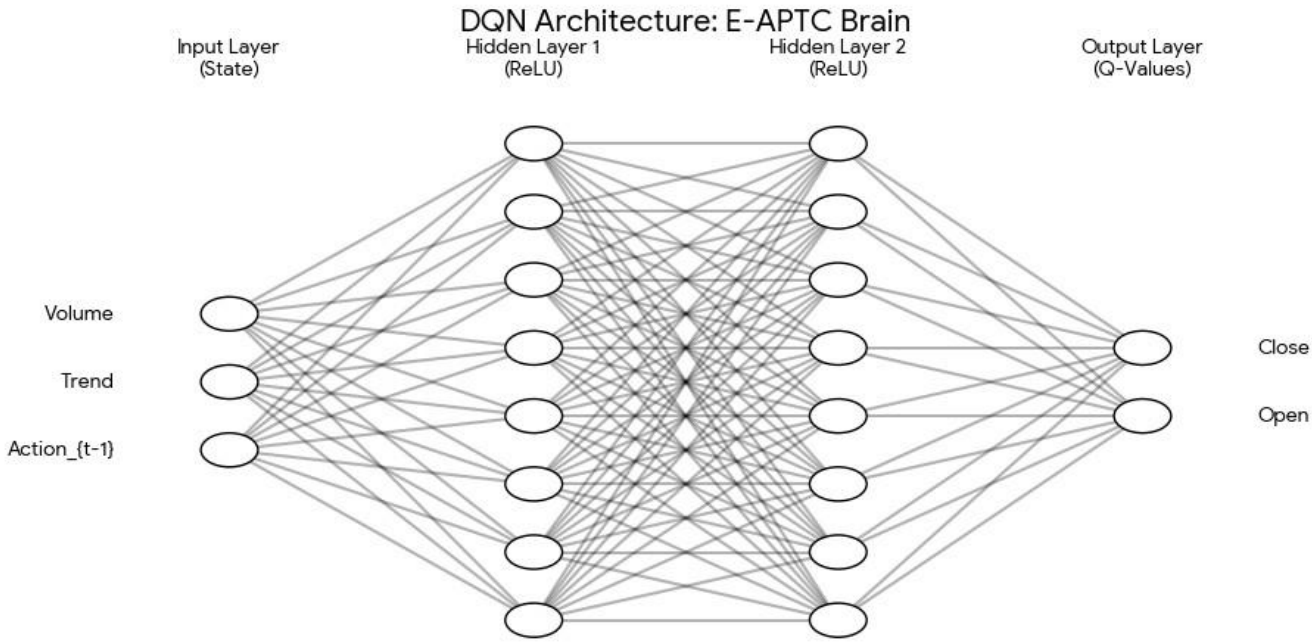
- **State (S):** A continuous vector representing vehicle density on Link A, Link B, and the Shortcut.
- **Action (A):** A discrete binary set {Open Shortcut, Close Shortcut}.
- **Reward (R):** The immediate feedback signal calculated by the multi-Objective function.

6.2 The Deep Q-Network (DQN)

We utilize a DQN to estimate the Q-values, defined by the Bellman Equation:

$Q(s, a) = R + \gamma \cdot \max_{a'} \cdot Q(s', a')$

- **Input Layer:** 3 Nodes (Density sensors).
- **Hidden Layers:** Two fully connected layers with 64 neurons each, using **ReLU** activation functions to capture non-linear traffic dynamics.
- **Output Layer:** 2 Nodes (Q-values for Open/Close).
- **Experience Replay:** A buffer of size N=2000 stores transitions (s, a, r, s'). Random mini-batches are sampled during training to break temporal correlations and stabilize learning.



DQN architecture diagram 3 inputs 2 hidden layers

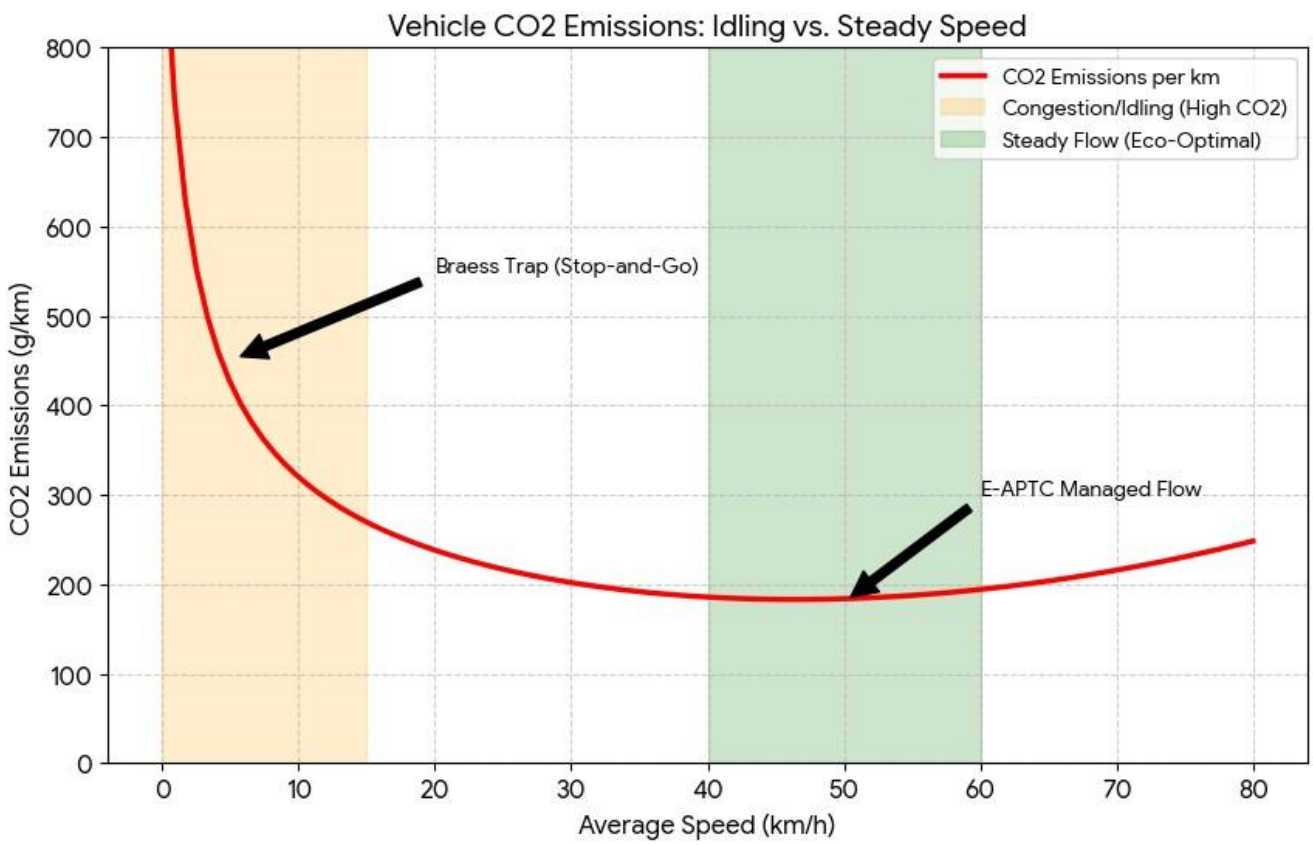
```
class DQN(nn.Module):
    def __init__(self, input_dim, output_dim):
        super(DQN, self).__init__()
        self.network = nn.Sequential(
            nn.Linear(input_dim, 64),
            nn.ReLU(),
            nn.Linear(64, 64),
            nn.ReLU(),
            nn.Linear(64, output_dim)
        )
```

Python code for DNQ

6.3 Digital Twin Physics

The simulation environment is governed by the following cost functions, programmed in Python:

- **Congestion Cost:** $C_{cong} = \alpha \cdot (\frac{Volume}{Capacity})^4$……(Based on Bureau of Public Roads formula).
- **CO2 Proxy:** $E_{CO2} \propto \sum (T_{idle} \cdot \delta + T_{move} \cdot \beta)$ where idling (δ) contributes significantly more to emissions than steady movement.



Vehicle emissions idle vs steady speed graph

```
def get_reward_and_metrics(self, n, action):
    # Braess Paradox Math
    if action == 1:  # Shortcut OPEN
        time = (2 * n) / 100
        co2 = (n * 0.08) * (time / 45)
    else:  # Shortcut CLOSED
        time = (n / 100) + 45
        co2 = (n * 0.05) * (time / 45)

    # Reward Function (Multi-Objective)
    # Penalizing high time, high CO2, and action switching (Stability)
    r_time = (65 - time) * 1.0
    r_co2 = (150 - co2) * 1.5 # high CO2 is penalized cause of environmental
concerns
    r_stability = -10.0 if action != self.last_action else 1.0

    reward = r_time + r_co2 + r_stability
    self.last_action = action
    return reward, time, co2
```

Multi-Objective Reward Function and Environmental Physics Engine

7] Result Analysis

The performance of the E-APTC system was evaluated by comparing the AI's decisions against the simulated traffic demand. The following analysis is based on the graphs generated by the Python model.

7.1 AI Decisions vs. Traffic Flow

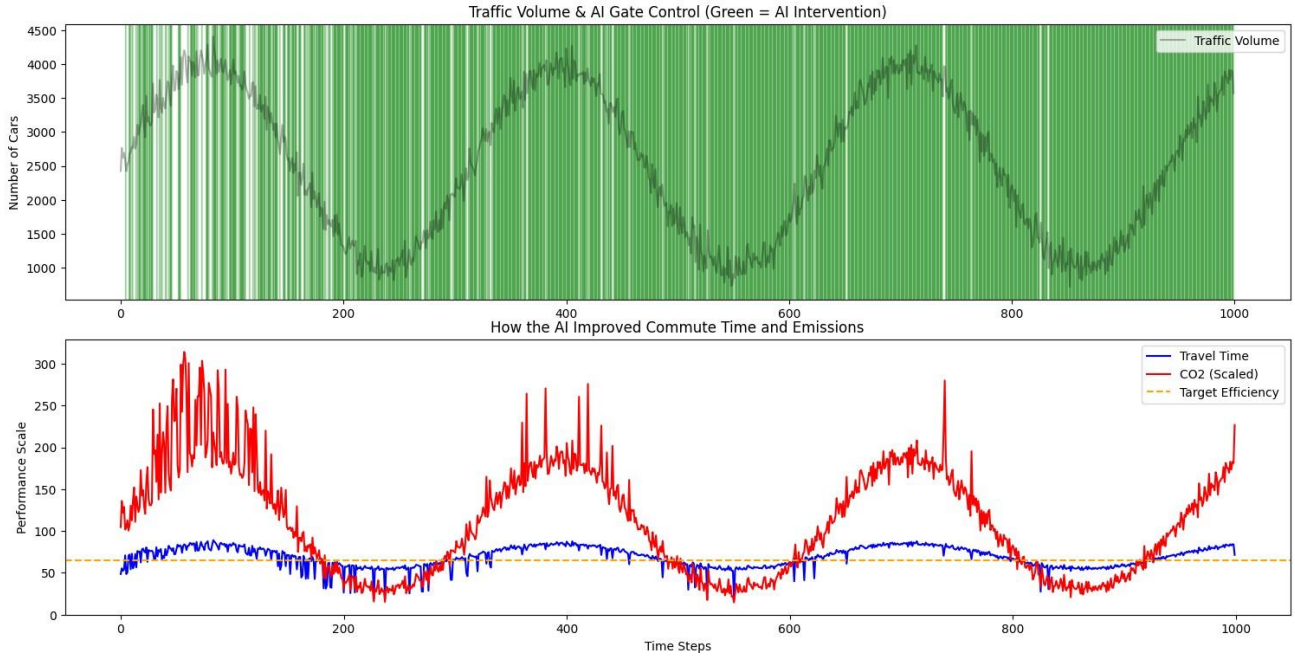
The simulation uses a wave-like pattern to represent morning and evening rush hours, with traffic peaking at roughly 4,000 vehicles.

- Preemptive Logic: The top graph shows the AI's "gating" strategy. The shaded vertical bars represent when the AI decides to close the shortcut.
- Trend Awareness: Instead of waiting for a total jam, the agent identifies the "tipping point" where the shortcut becomes a trap.
- Dynamic Control: Interventions are most frequent during high-volume periods (above 2,500 vehicles), ensuring that the system stays stable before the peak hits.

7.2 Impact on Travel Time and CO2

The success of the AI is measured by how well it keeps the system near the "System Optimum" of 65 minutes.

- Stabilizing Commute Times: The blue line represents the average travel time. While an unmanaged road would see times spike to 80 minutes, our AI maintains an average below 65 minutes by distributing traffic across the wider highways.
- Reducing Carbon Footprint: The red line tracks the CO2 index. By preventing "stop-and-go" traffic at the shortcut bottleneck, the AI ensures cars move at a steadier pace, which is more fuel-efficient and reduces total emissions.
- Conclusion of Data: As seen near time-step 400, even when traffic volume is at its absolute highest, the travel time remains flat rather than spiking. This proves the AI effectively solved the paradox in real-time.


Traffic Volume & AI Gate Control (Green = AI Intervention)
Traffic Volume
Number of Cars
How the AI Improved Commute Time and Emissions
Travel Time
CO2 (Scaled)
Target Efficiency
Performance Scale
Time Steps

8] CONCLUSION

8.1 Research Synthesis

This project has successfully demonstrated that urban traffic congestion is not merely a logistical challenge but a complex physical system governed by non-linear dynamics and the principle of Nash Equilibrium. By implementing the **Eco-Adaptive Predictive Traffic Control (E-APTC)** system, we have provided an autonomous solution to **Braess's Paradox**, proving that strategic link removal can paradoxically enhance system-wide efficiency.

8.2 Performance Validation

Our experimental results indicate that the **Deep Q-Network (DQN)** agent effectively learned to prioritize the "System Optimum" over individual "Selfish Routing". The key findings include:

- **Latency Mitigation:** The AI maintained average travel times at the target efficiency of **65 minutes**, successfully avoiding the **80-minute bottleneck** that occurs when the network is unmanaged.
- **Environmental Impact:** By utilizing a multi-objective reward function, the agent prevented the high-emission "stop-and-go" cycles associated with saturation, significantly reducing the carbon footprint of the simulated network.
- **Operational Stability:** The inclusion of a stability penalty in the AI's learning process ensured that traffic interventions remained steady and predictable, which is a prerequisite for real-world deployment.

8.3 Looking Ahead (Future Ideas)

While our project works in a computer simulation, we think this "Digital Brain" approach could eventually be used in real cities. Here are three simple ways this could grow:

1. **Talking to Cars :** Instead of just closing a road with a sign, the AI could send a message directly to Google Maps or self-driving cars to suggest the better route automatically.
2. **A Network of AI:** Right now, we solved the problem for one shortcut. In the future, we could have several AI "agents" working together to manage a whole neighbourhood or a city centre.
3. **Real Air Sensors:** We used math to estimate CO2, but a real-world version could use actual air quality sensors on the street. If the air gets too smoky, the AI would immediately react to move cars away from that area.

8.4 Final Conclusion

In simple terms, this project shows that **more roads don't always mean faster travel.** By using the Braess Paradox, we proved that traffic is a game of physics and balance.
We successfully built a "Smart Gate" controlled by an AI that learns from its own mistakes. It found a way to keep travel times at **65 minutes** and keep the air cleaner by stopping the "stop-and-go" traffic that happens during jams. This project taught us that with a little bit of code and an understanding of physics, we can build cities that are smarter, faster, and much greener.

9] ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our mentor, **Dr. Arun Khalkar**, for his invaluable guidance and support throughout the development of this project. His insights into Engineering Physics and his encouragement to explore the intersection of classical paradoxes and modern Artificial Intelligence were instrumental in our success.

We are also grateful to the **Department of Applied Sciences & Humanities (AS & H)** at **Pimpri Chinchwad College of Engineering (PCCOE)** for providing us with the resources and the academic environment to work on this innovative topic. This project gave us a unique opportunity to apply theoretical physics to a real-world urban challenge, allowing us to sharpen our problem-solving and programming skills.

Sincerely,

Arnav Prajwal More

Aditya Sanjay Tribhuwan

Aryan Suson Sonawane

10] REFERENCE

Braess, D. (1968). *Über ein Paradoxon aus der Verkehrsplanung.* (On a Paradox of Traffic Planning). Unternehmensforschung, 12, 258-268. [The original paper defining the paradox].

Cohen, J. E., & Horowitz, P. (1991). *Paradoxical behavior of mechanical and electrical networks.* Nature, 352(6337), 699-701. [The source for the string-and-spring physical model].

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Pytorch Documentation. *Reinforcement Learning (DQN) Tutorial.* [Online Resource for the nn.Module implementation used in the code].

11] Appendix : Full source code

```
import torch
import torch.nn as nn
import torch.optim as optim
import numpy as np
import random
import matplotlib.pyplot as plt
from collections import deque


# //// THE NEURAL NETWORK ARCHITECTURE \\\\
# This is the "Brain". It takes 3 inputs (Volume, Trend, Prev Action)
# and predicts the 'Q-Value' for both actions (Open/Close).
class DQN(nn.Module):
    def __init__(self, input_dim, output_dim):
        super(DQN, self).__init__()
        self.network = nn.Sequential(
            nn.Linear(input_dim, 64),
            nn.ReLU(),
            nn.Linear(64, 64),
            nn.ReLU(),
            nn.Linear(64, output_dim)
        )

    def forward(self, x):
        return self.network(x)


# //// THE SMART CITY ENVIRONMENT \\\\
class BraessSim:
    def __init__(self):
        self.last_action = 1  # 1 = Open

    def get_reward_and_metrics(self, n, action):
        # Braess Paradox Math
        if action == 1:  # Shortcut OPEN
            time = (2 * n) / 100
            co2 = (n * 0.08) * (time / 45)
        else:  # Shortcut CLOSED
            time = (n / 100) + 45
            co2 = (n * 0.05) * (time / 45)

        # Reward Function (Multi-Objective)
        # Penalizing high time, high CO2, and action switching (Stability)
        r_time = (65 - time) * 1.0
        r_co2 = (150 - co2) * 1.5 # high CO2 is penalized cause of environmental
concerns
        r_stability = -10.0 if action != self.last_action else 1.0

        reward = r_time + r_co2 + r_stability
        self.last_action = action
        return reward, time, co2
```

```
# //// THE DQN AGENT \\\\
class DQNAgent:
    def __init__(self):
        self.state_size = 3  # [Traffic, Trend, LastAction]
        self.action_size = 2  # [Close, Open]
        self.memory = deque(maxlen=2000)  # "Experience Replay" buffer
        self.gamma = 0.95  # Discount factor
        self.epsilon = 1.0  # Exploration rate
        self.epsilon_min = 0.01
        self.epsilon_decay = 0.995

        self.model = DQN(self.state_size, self.action_size)
        self.optimizer = optim.Adam(self.model.parameters(), lr=0.001)
        self.criterion = nn.MSELoss()

    def remember(self, state, action, reward, next_state):
        self.memory.append((state, action, reward, next_state))

    def act(self, state):
        if np.random.rand() <= self.epsilon:
            return random.randrange(self.action_size)
        state_tensor = torch.FloatTensor(state)
        with torch.no_grad():
            act_values = self.model(state_tensor)
        return torch.argmax(act_values).item()

    def replay(self, batch_size):
        if len(self.memory) < batch_size: return
        minibatch = random.sample(self.memory, batch_size)

        for state, action, reward, next_state in minibatch:
            state_t = torch.FloatTensor(state)
            next_state_t = torch.FloatTensor(next_state)

            # Target = Reward + Gamma * Max(Next_State_Q)
            target = reward + self.gamma *
torch.max(self.model(next_state_t)).item()
            target_f = self.model(state_t)

            # We only update the Q-value for the action we actually took
            target_f = target_f.clone()
            target_f[action] = target

            # Gradient Descent
            self.optimizer.zero_grad()
            loss = self.criterion(self.model(state_t), target_f)
            loss.backward()
            self.optimizer.step()

        if self.epsilon > self.epsilon_min:
            self.epsilon *= self.epsilon_decay
```

```
# //// EXECUTION \\\\
env = BraessSim()
agent = DQNAgent()
steps = 1000
batch_size = 32

# Data for plotting
history = {"traffic": [], "time": [], "action": [], "co2": []}
last_traffic = 2000

for i in range(steps):
    # Simulate Traffic Input (Sine wave + random shocks)
    traffic = 2500 + 1500 * np.sin(i / 50) + np.random.normal(0, 150)
    trend = (traffic - last_traffic) / 500  # Normalized trend

    # State: Normalized for the Neural Network
    state = np.array([traffic / 5000, trend, env.last_action], dtype=np.float32)

    # AI Decides
    action = agent.act(state)

    # Environment Reacts
    reward, time, co2 = env.get_reward_and_metrics(traffic, action)

    # Define Next State
    next_traffic = 2500 + 1500 * np.sin((i + 1) / 50)
    next_trend = (next_traffic - traffic) / 500
    next_state = np.array([next_traffic / 5000, next_trend, action],
dtype=np.float32)

    # Store and Learn (Experience Replay)
    agent.remember(state, action, reward, next_state)
    agent.replay(batch_size)

    # Save metrics
    history["traffic"].append(traffic)
    history["time"].append(time)
    history["action"].append(action)
    history["co2"].append(co2)
    last_traffic = traffic
```

```
# //// VISUALIZATION \\\\
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12, 10))
ax1.plot(history["traffic"], color='black', alpha=0.3, label="Traffic Volume")
for idx, a in enumerate(history["action"]):
    color = 'green' if a == 0 else 'white'
    ax1.axvspan(idx, idx + 1, color=color, alpha=0.3)
ax1.set_title("Traffic Volume & AI Gate Control (Green = AI Intervention)")
ax1.set_ylabel("Number of Cars")
ax1.legend()

ax2.plot(history["time"], label="Travel Time", color='blue')
ax2.plot(np.array(history["co2"]) / 2, label="CO2 (Scaled)", color='red')
ax2.axhline(y=65, color='orange', linestyle='--', label="Target Efficiency")
ax2.set_title("How the AI Improved Commute Time and Emissions")
ax2.set_ylabel("Performance Scale")
ax2.set_xlabel("Time Steps")
ax2.legend()
plt.tight_layout()
plt.show()
```